Sreeram Ganesan

linkedin.com/in/sreeramganesan
github.com/sreeram-gsan

Website: www.sreeramganesan.com
Email: srga8641@colorado.edu
Mobile: 720-803-8838

Skills

Languages: Python, Java, JavaScript, SQL, HTML, CSS

Frameworks & Libraries: Spring Boot, Flask, Node.js, React, gRPC, RabbitMQ, Kafka

Databases: MySQL, PostgreSQL, Oracle, MongoDB, DynamoDB, Redis, Cassandra, Cloud Spanner, BigQuery, Aurora, Apache Hudi, SQLite (FTS5, sqlite-vec)

Cloud & Infrastructure: AWS (S3, CDK, Bedrock, SageMaker, etc), Google Cloud Platform, Oracle Cloud Infrastructure, Docker, Kubernetes, Karpenter, KServe, Terraform, Jenkins, CI/CD

Systems & Data: Linux, Spark (Spark SQL), Hadoop, Apache OpenWhisk, OAuth, Load Balancing, RESTful APIs, Microservices

ML & AI: PyTorch, LangChain, RAG, MCP, LLM Agents, Embedding & Reranker Models, Hybrid & Vector Search, Entity Resolution, Knowledge Graphs, vLLM, SGLang, Hugging Face TEI, MLX, Quantization, Ollama, MLflow, Ray, Fine-tuning

Dev Tools: Kiro, Claude Code, Git, Mercurial, Gradle, Jetty, JDK

Experience

Machine Learning Engineer II — June 2025 - Present
Amazon AGI — Sunnyvale, United States

- Developing large-scale knowledge graphs and agentic search platform that gives AI agents structured, grounded retrieval over web-scale data, serving as the data foundation for LLM grounding and training.
- Built a hybrid entity resolution pipeline: Spark SQL rule-based blocking extended by LLM agents that analyze data and quality metrics to author new blocking rules, plus semantic-search blocking over per-class vector indexes, a BGE reranker to prune candidate pairs, and a Qwen 3.5 4B LLM matcher; improved recall by 11% and precision by 20% while cutting LLM matcher calls by 92.2%.
- Built AKG ONIX (One Inference Experience), a scalable inference control plane using Karpenter and KServe that autoscales model serving across GPU instance pools, serving Qwen 3.5 via vLLM and the reranker via Hugging Face TEI; reduced GPU cost by 35% and sustained 50 req/s at p95 latency (for Qwen matcher) under 1500 ms.
- Built a streaming vectorization pipeline that continuously indexes and re-indexes 3 billion+ documents as they change, keeping the embedding index fresh and feeding hybrid (vector + keyword) search for low-latency, high-precision retrieval.

Software Engineer II — Jun 2023 - May 2025
Amazon Web Services — Palo Alto, United States

- Optimized the canary testing methodology for the Aurora MySQL engine, generating annual cost savings of $16.5 million for AWS. Served as an Aurora representative and showcased this optimization initiative to SVPs during AWS-wide operations review.
- Designed and implemented a sophisticated control plane monitoring system that proactively identifies critical issues and automatically diagnoses their root causes, thereby saving significant engineering time.
- Designed and developed InfraGPT, an internal LLM which used RAG to automatically triage issues and failures. InfraGPT was single handedly responsible for reducing ops queries to the Infra team by 56%. Engineered a Model Context Protocol server that enhanced InfraGPT's capabilities, enabling it to perform basic infrastructure maintenance operations such as recovering leaked resources.
- Served as a Subject Matter Expert (SME) for the Infrastructure team while mentoring 8 engineers and 3 interns, providing technical guidance. Developed training programs and conducted regular knowledge-sharing sessions that accelerated team members' technical competency and project readiness.

Software Engineering Intern — Aug 2022 - Dec 2022
Google — Boulder, United States

- Designed, developed and released an automated schema roll-out framework which enables Google Cloud and its partners to modify the schema of existing billing exports without any data copy/duplication.
- Automated the feature roll-out process for 170+ Google Cloud partners, reducing the time to roll-out new features from 4 weeks (done manually) to less than a week (completely automated).

Software Engineering Intern — May 2022 - Aug 2022
Amazon Web Services — Palo Alto, United States

- Architected and engineered an internal tool to automatically validate and rank the Standard Operating Procedures (SOP) for fixing production issues in the Aurora database using AWS services like Lambda, SNS, SQS and DynamoDB.
- Significantly reduced the time taken to analyze, identify and review SOPs which are crucial for maintaining Aurora in production.

Software Engineer Jan 2018 - Jul 2021
Oracle Hyderabad, India

- Achieved 3X performance gain by developing new microservices with Java to import and export large amounts of data via Excel and other third-party apps. Reduced server cost by 34% compared to the legacy implementation.
- Led a team of three to prototype and construct a novel revenue forecasting solution. Worked closely with product managers to incorporate customer feedback and improved the forecast by 13%. Promoted from Associate Software Engineer to Software Engineer in less than a year.
- Created RESTful APIs to configure, generate and export mission-critical reports for R&A, a data analytics tool used by restaurant chains such as Starbucks, Dunkin, KFC, etc to optimize sales, operation costs, inventory, etc.
- Decreased the turnaround time for critical bugs by 41% by re-architecting the legacy modules and their build process.
- Devised a proof of concept for an internal framework to build mobile applications with the web development stack - HTML, CSS and Javascript, which cut down the development time of mobile apps by 45%.

Education

Master of Science in Computer Science May 2023
University of Colorado Boulder, Boulder, CO; GPA: 3.92/4

Deep Learning, Natural Language Processing, Data Mining, Datacenter-Scale Computing, Big Data Architecture

Bachelor of Technology in Computer Science Jul 2018
Amrita School of Engineering, Amrita University, Coimbatore, India; GPA: 3.53/4

Relevant Coursework: Data Mining and Data Warehousing, Big Data Analytics, Machine Learning

Projects

Dowse - On-Device Semantic Search for macOS Oct 2026

- Built a macOS native launcher and CLI that indexes local files (PDFs, office docs, code, notes, images via OCR) and retrieve them by meaning using Embedding Gemma 2 running fully on device with no network calls.
- Supports ncremental indexing via FSEvents and content-hash diffing indexes and reembeds edited files at low QoS.
- Implemented hybrid retrieval fusing dense vectors with SQLite FTS5 BM25 through Reciprocal Rank Fusion, lifting Hit@5 from 58% (keyword-only) to 82% on a labeled evaluation harness; Matryoshka truncation and int8 quantization shrank the index 12x with under 2% recall loss at 45 ms p50 query latency.
- Tech Stack - Swift, SwiftUI, Embedding Gemma 2, MLX/Core ML, SQLite (sqlite-vec, FTS5), FSEvents, Vision OCR, XPC

Accelerating the communication between Serverless functions using RDMA May 2023

- Serverless computing is mainly targeted for stateless functions that do minimal I/O and communication. The existing solutions to use Serverless computing for Stateful functions are not very efficient.
- This project achieves high performance inter-function communication by co-locating the functions on the same VM or by using RDMA when the functions are located on different VMs.
- Tech Stack - Python, Shell scripting, Docker, Kubernetes, Apache OpenWhisk

F7 Fitness Hub - Social Fitness Network Jul 2022

- Architected and developed mobile applications for a medium-sized gym to build a community for their clients to virtually participate in various workouts from their mobile.
- Responsible for supporting clients and generating 100% of the revenue for the business during COVID-19 pandemic lockdowns.
- Tech Stack - Java, Spring Boot, MySQL, React Native.

Revenue Forecasting with Machine Learning May 2021

- Engineered a system to forecast the revenue of restaurants with Machine Learning. Engineered an ExtraTrees Regressor-based ML model that predicted the weekly revenue of restaurants with a mean absolute error of less than 15%.
- Tech Stack - Python, Flask, Tensorflow, Oracle JET, Oracle Database.

Achievements

Felicitated by the Minister of Rural Development, Government of India - Established a beneficiary management application for a government program. Constructed dashboards and reports to measure the impact of the program.

Inspiration Award - Awarded by Persistent Systems for winning the Smart India Hackathon.

Second Runner Up, Honeywell Hackathon - Developed a chrome extension for personalised product recommendation.